

1. Student Roll Number Search

A teacher has stored the roll numbers of students in a list in the order they registered. The list is not sorted.

Write a program to check whether a given roll number exists in the list. If found, display its position; otherwise, print "Student Not Found."

2. Product ID Search

An e-commerce website stores product IDs in ascending order.

Write a program to find whether a customer-entered product ID exists in the inventory. If it exists, display its index; otherwise, display "Product Not Available."

3. Arrange Exam Marks

A teacher wants to display students' marks from the lowest to the highest.

Write a program to sort the marks of all students in ascending order.

4. Rank Participants

A sports academy has recorded the timings (in seconds) of participants in a race.

Write a program to arrange the timings from the fastest to the slowest so that the winners can be announced.

5. Insert New Book by Price

A bookstore maintains a sorted list of book prices.

A new book arrives, and its price needs to be placed at the correct position while keeping the list sorted.

Write a program to perform this task.

6. Employee Salary Report

An HR department has employee salary records collected from two different branches.

Write a program to combine both lists and display all salaries in ascending order.

7. Online Game Leaderboard

An online gaming platform stores players' scores.

Write a program to arrange the scores in descending order so that the leaderboard can be displayed.

8. Hospital Emergency Queue

A hospital has a list of patients with different priority levels.

Write a program to arrange the patients so that the patient with the highest priority is treated first.

9. Library Book Search

A library has 10,000 books, and the Book IDs are already arranged in ascending order.

Write a program to find a given Book ID efficiently.

Also mention which searching algorithm you used and why it is suitable.

10. School Annual Report

A school has recorded the marks of 50 students.

Write a program that:
Sorts the marks in ascending order.
Accepts a mark from the user.
Checks whether that mark exists in the sorted list.
Displays the position if found; otherwise, prints "Mark Not Found."